

Brief: Mastering the Data Dashboard

Module: Data Visualization and Analytics

Task: Presentation

Schedule: To be worked on Tuesday afternoon and Wednesday

Due: Meetings will be held on Teams on Thursday (30th) afternoon to report findings

1. Project Overview

In an era of "data overload," the ability to distill complex datasets into actionable insights is a critical skill. This assignment challenges you to move beyond simple charting and explore the **psychology and engineering** of high-performance dashboards. You are tasked with researching, designing, and presenting a dashboard concept that adheres to industry-standard visual perception "laws".

In your presentation of your findings you must also provide the rationale for the existence of dashboards, who they are useful for and what purpose they serve.

2. Core Objectives

Your presentation must address the following three pillars of dashboard design:

A. The Fundamentals of Visual Perception

Explain how human cognition influences design. You should research:

- **Gestalt Principles:** How proximity, similarity, and enclosure group data points.
- **Preattentive Attributes:** Using color, size, and orientation to direct the user's eye.
- **Data-Ink Ratio:** demonstrate how to strip away non-essential "chart junk."

B. Functional Hierarchy & Layout

A dashboard is not just a collection of widgets; it tells a story. Your brief must cover:

- **The Inverted Pyramid:** Placing Key Performance Indicators (KPIs) at the top, followed by trends, and ending with granular details.
- **The F-Pattern:** Designing for how users naturally scan screens.

C. Technical Constraints & Interactivity

- How do we balance **real-time data** vs. **historical snapshots**?
 - The importance of drill-down capabilities and filters without cluttering the UI.
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3. Deliverables

1. **A 10-Minute Presentation:** Explaining your findings and the design logic used to serve them.
2. **The Mockup:** Apply the principles you have learned about and create a static mock up of the plots you produced on Monday **in powerpoint**, detailing what interactive elements you would like to apply and how you would like to apply them. I.e you may choose to show ridership over time but you might want to add a filtering option to look between specific dates, regions or modes of transport.

If your mock “dashboard” would be made of multiple tabs you can display these as different slides.
